

The Colonial Origins of Comparative Development: An Empirical Investigation

Daron Acemoglu, Simon Johnson and James A. Robinson
(AER 2001)

Presented by Hanna Schwank

EC 765

April 2, 2018

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

"The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data." (Albouy 2004b)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

"The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data." (Albouy 2004b)

"Response to Albouy's 'A Reexamination Based on Improved Settler Mortality Data.'" (AJR 2005)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

"The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data." (Albouy 2004b)

"Response to Albouy's 'A Reexamination Based on Improved Settler Mortality Data.'" (AJR 2005)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2006)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

"The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data." (Albouy 2004b)

"Response to Albouy's 'A Reexamination Based on Improved Settler Mortality Data.'" (AJR 2005)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2006)

"Reply to the Revised (May 2006) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2006)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

"The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data." (Albouy 2004b)

"Response to Albouy's 'A Reexamination Based on Improved Settler Mortality Data.'" (AJR 2005)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2006)

"Reply to the Revised (May 2006) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2006)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2008)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

"The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data." (Albouy 2004b)

"Response to Albouy's 'A Reexamination Based on Improved Settler Mortality Data.'" (AJR 2005)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2006)

"Reply to the Revised (May 2006) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2006)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2008)

"Reply to the Revised (2008) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2008)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

"The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data." (Albouy 2004b)

"Response to Albouy's 'A Reexamination Based on Improved Settler Mortality Data.'" (AJR 2005)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2006)

"Reply to the Revised (May 2006) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2006)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2008)

"Reply to the Revised (2008) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2008)

"The Colonial Origins of Comparative Development: An Empirical Investigation: Comment (Albouy 2012)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

"The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data." (Albouy 2004b)

"Response to Albouy's 'A Reexamination Based on Improved Settler Mortality Data.'" (AJR 2005)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2006)

"Reply to the Revised (May 2006) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2006)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2008)

"Reply to the Revised (2008) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2008)

"The Colonial Origins of Comparative Development: An Empirical Investigation: Comment (Albouy 2012)

The Colonial Origins of Comparative Development: An Empirical Investigation: Reply (AJR 2012)

A Series of Papers...

"The Colonial Origins of Comparative Development: An Empirical Investigation" (AJR 2001)

"The Colonial Origins of Comparative Development: A Reinvestigation of the Data." (Albouy 2004a)

"The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data." (Albouy 2004b)

"Response to Albouy's 'A Reexamination Based on Improved Settler Mortality Data.'" (AJR 2005)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2006)

"Reply to the Revised (May 2006) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2006)

"The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data." (Albouy 2008)

"Reply to the Revised (2008) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.'" (AJR 2008)

"The Colonial Origins of Comparative Development: An Empirical Investigation: Comment (Albouy 2012)

The Colonial Origins of Comparative Development: An Empirical Investigation: Reply (AJR 2012)

Acemoglu, D., Johnson, S., & Robinson, J. A. (2001). The colonial origins of comparative development: An empirical investigation. *American Economic Review*, 91(5), 1369-1401.

Motivation

Want to know the effect of institutions (property rights enforcement, constraints on government expropriation, institutions providing equal access to education, ...) on income per capita.

Problems:

- ▶ Reverse causality
- ▶ Omitted variables

→ Need exogenous variation in institutions.

Instrument: (Potential) Settler Mortality

Instrument: mortality rates expected by the first European settlers in the colonies

(potential) settler mortality → settlements → early institutions → current institutions → current performance

Three Premises

1. There were different types of colonization policies which created different sets of institutions.
2. The colonization strategy was influenced by the feasibility of settlements.
3. The colonial state and institutions persisted even after independence.

Types of Colonization and Settlements

- ▶ "Neo-Europes"
 - ▶ Canada, Australia, US, New Zealand
 - ▶ Representative institutions very similar to Europe
 - ▶ Settlers demanding equal rights
- ▶ Extractive institutions
 - ▶ Slave trade in Africa
 - ▶ Natural resources like gold from Latin America and Africa

Mortality of Early Settlers

- ▶ Malaria and yellow fever
 - ▶ 80% of European deaths in the tropics
 - ▶ Local population has immunity (malaria is most serious for young children)
 - ▶ Europeans started to control these diseases in the late 19th century
- ▶ Gastrointestinal diseases
 - ▶ 15% of European deaths in the tropics
 - ▶ Depends on living situation

Institutional Persistence

Young (1994, pg. 283):

"although we commonly described the independent polities as 'new states,' in reality they were successors to the colonial regime, inheriting its structures, its quotidian routines and practices, and its more hidden normative theories of governance"

3 possible mechanisms:

1. Setting up institutions that place restrictions on government power and enforce property rights is costly
2. Gains to extractive institutions are bigger the smaller the size of the ruling elite
3. If agents make irreversible investments complementary to a particular set of institutions, they will be more willing to support them

Data: Potential Settler Mortality

- ▶ Data on mortality of European troops during the 19th century by Curtin
 - ▶ Pick earliest available data
 - ▶ Assign mortality rates to neighboring countries with similar disease environment
- ▶ South America: (almost) no soldier data available, use data on mortality rates of bishops in Latin America from 1604 to 1876 (scaled up)
- ▶ Mortality rates range from 8.55 (Australia and New Zealand) to 2940 (Mali) annualized deaths among 1,000 soldiers where each death is replaced with a new soldier

Data: Institutions and Economic Performance

- ▶ Performance: log GDP per capita (PPP) in 1995 or log output per worker in 1988
- ▶ Current institutions: average protection against expropriation risk in 1985-1995 or constraints on executive in 1990
- ▶ Early institutions: constraints on executive in 1900 or in first year of independence, democracy in 1900

Sample includes 64 former colonies.

Institutions and Performance: OLS Estimates

$$\log y_i = \mu + \alpha R_i + X_i' \gamma + \epsilon_i \quad (1)$$

y_i : GDP per capita

R_i : protection against expropriation measure

X_i : controls

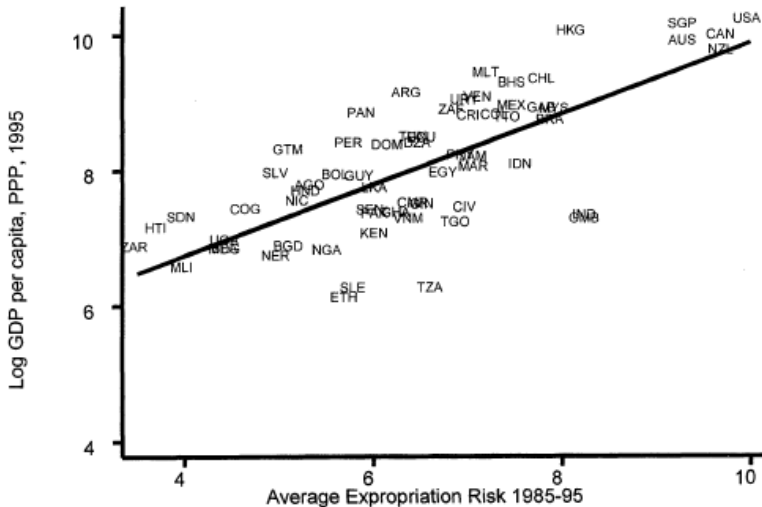


Figure 1: OLS Relationship Between Institutions and Development

Institutions and Performance: IV Regression

$$R_i = \lambda_R + \beta_R C_i + X_i' \gamma_R + \nu_{R_i} \quad (2)$$

$$C_i = \lambda_C + \beta_C S_i + X_i' \gamma_C + \nu_{C_i} \quad (3)$$

$$S_i = \lambda_S + \beta_S \log M_i + X_i' \gamma_S + \nu_{S_i} \quad (4)$$

C_i : Early institutions

S_i : Settlements

M_i : (Potential) settler mortality

Panel A

Dependent Variable Is Average Protection Against Expropriation Risk in 1985–1995

Constraint on executive in 1900	0.32 (0.08)	0.26 (0.09)								
Democracy in 1900			0.24 (0.06)	0.21 (0.07)						
Constraint on executive in first year of independence					0.25 (0.08)	0.22 (0.08)				
European settlements in 1900							3.20 (0.61)	3.00 (0.78)		
Log European settler mortality									-0.61 (0.13)	-0.51 (0.14)
Latitude		2.20 (1.40)		1.60 (1.50)		2.70 (1.40)		0.58 (1.51)		2.00 (1.34)
R^2	0.2	0.23	0.24	0.25	0.19	0.24	0.3	0.3	0.27	0.3
Number of observations	63	63	62	62	63	63	66	66	64	64

(2)

Figure 2: Relationship Between Early Institutions and Current Institutions

Panel B	Dependent Variable Is Constraint on Executive in 1900				Dependent Variable Is Democracy in 1900				Dependent Variable Is European Settlements in 1900	
European settlements in 1900	5.50 (0.73)	5.40 (0.93)			8.60 (0.90)	8.10 (1.20)	(3)		(4)	
Log European settler mortality			-0.82 (0.17)	-0.65 (0.18)			-1.22 (0.24)	-0.88 (0.25)	-0.11 (0.02)	-0.07 (0.02)
Latitude		0.33 (1.80)		3.60 (1.70)		1.60 (2.30)		7.60 (2.40)		0.87 (0.19)
R ²	0.46	0.46	0.25	0.29	0.57	0.57	0.28	0.37	0.31	0.47
Number of observations	70	70	75	75	67	67	68	68	73	73

Figure 3: Relationship Between Mortality, Settlers, and Early Institutions

Two Stages Least Squares Estimation

First Stage:

$$R_i = \xi + \beta \log M_i + X_i' \delta + \nu_i \quad (5)$$

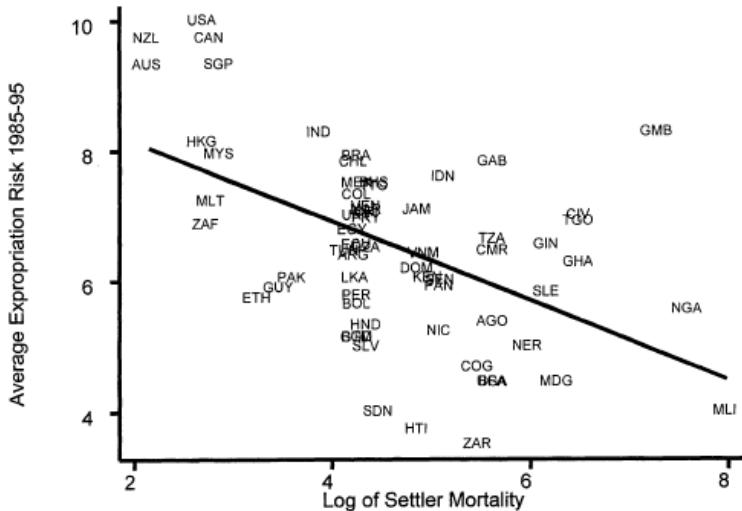


Figure 4: First Stage Relationship between Log Mortality and Index of Institutions

	Base sample (1)	Base sample (2)	Base sample without Neo-Europes (3)	Base sample without Neo-Europes (4)	Base sample without Africa (5)	Base sample without Africa (6)	Base sample with continent dummies (7)	Base sample with continent dummies (8)	Base sample, dependent variable is log output per worker (9)
Panel A: Two-Stage Least Squares									
Average protection against expropriation risk 1985–1995	0.94 (0.16)	1.00 (0.22)	1.28 (0.36)	1.21 (0.35)	0.58 (0.10)	0.58 (0.12)	0.98 (0.30)	1.10 (0.46)	0.98 (0.17)
Latitude		−0.65 (1.34)		0.94 (1.46)		0.04 (0.84)		−1.20 (1.8)	
Asia dummy							−0.92 (0.40)	−1.10 (0.52)	
Africa dummy							−0.46 (0.36)	−0.44 (0.42)	
“Other” continent dummy							−0.94 (0.85)	−0.99 (1.0)	
Panel B: First Stage for Average Protection Against Expropriation Risk in 1985–1995									
Log European settler mortality	−0.61 (0.13)	−0.51 (0.14)	−0.39 (0.13)	−0.39 (0.14)	−1.20 (0.22)	−1.10 (0.24)	−0.43 (0.17)	−0.34 (0.18)	−0.63 (0.13)
Latitude		2.00 (1.34)		−0.11 (1.50)		0.99 (1.43)		2.00 (1.40)	
Asia dummy							0.33 (0.49)	0.47 (0.50)	
Africa dummy							−0.27 (0.41)	−0.26 (0.41)	
“Other” continent dummy							1.24 (0.84)	1.1 (0.84)	
R^2	0.27	0.30	0.13	0.13	0.47	0.47	0.30	0.33	0.28

Figure 5: IV Regression Results: Effect of Institutions on Economic Growth

Additional Robustness Checks

- ▶ Identity of colonial power: Dummies for British and French colonies
- ▶ Estimation for British colonies only
- ▶ Legal origin: French legal origin dummy
- ▶ Religion: Fraction population being Catholic, Muslim, or of other religions
- ▶ Climate: Control for temperature and humidity
- ▶ European culture: Fraction of population of European descent
- ▶ Geography: Natural resources, soil type, landlocked
- ▶ Ethnolinguistic fragmentation (endogenous?)
- ▶ Disease Environment: Fraction of population living in a malaria area (endogenous)
- ▶ Health: Life expectancy and infant mortality (highly endogenous)

Albouy, D. Y. (2012). The colonial origins of comparative development: an empirical investigation: comment. *American Economic Review*, 102(6), 3059-76.

and

Acemoglu, D., Johnson, S., & Robinson, J. A. (2012). The colonial origins of comparative development: An empirical investigation: Reply. *American Economic Review*, 102(6), 3077-3110.

Albouy's Main Critiques

- ▶ Only 28 out of 64 countries have mortality rates that originate from within their own borders - the others are assigned by AJR from countries with similar disease environments
→ drops them
 - ▶ 16 extrapolated from Latin American bishop data using a benchmarking procedure
 - ▶ Six assignments based on incorrect interpretation of former colonial names for Mali
- ▶ Mortality rates never come from European settlers, but from soldiers
 - ▶ Sometimes soldiers at peace in barracks
 - ▶ Some others on campaign
- ▶ Use of "conventional" standard errors, no clustering or weak instruments robust confidence intervals

Benchmarking Latin American Data

Albouy:

- ▶ AJR use Gutierrez (1986) data to assign mortality rates to 16 Latin American countries
- ▶ Gutierrez categorizes cities with bishops into low-, medium-, and high-temperature regions - rates are based on 4,5, and 10 deaths out of at-risk populations of 24, 28.5, and 30.5 bishops in each region over 10 years
- ▶ AJR multiply resulting rates by 4.25 to benchmark them to a mortality rate in Mexico (where Curtin and Gutierrez data are available)
 - ▶ Depending on benchmarking procedure, ratio of soldier to bishop mortality rates range from 0.98 to 10.8

Benchmarking Latin American Data

AJR:

- ▶ Use Jamaica/Dominican Republic as alternative benchmarking
- ▶ Use data from British naval stations in modern Argentina, Brazil, Chile, Peru, and Panama
 - ▶ Drop all other Latin American countries
- ▶ Additionally, use life insurance rates for sailors on the "South American Station"

Matching Mortality Rates to Neighboring Countries

Albouy:

- ▶ AJR assign mortality number to a country if it neighbors a country for which they have data and that has same disease environment
- ▶ In sub-Saharan Africa and Southeast Asia, neighboring countries in the data have very different rates
 - ▶ Assignment very sensitive to choice of neighbor
- ▶ Resulting mortality rates are too high

Matching Mortality Rates to Neighboring Countries

AJR:

- ▶ Some mortality rates are very high because they come from epidemics
 - ▶ Use log mortality to reduce influence of outliers
 - ▶ Alternative: cap mortality rates at 250 (150, 280, 350) since everything above probably didn't further influence settlement decisions
- ▶ Capping is preferred to trimming outliers, because it preserves information and the sample is already small

Other Countries

- ▶ Albouy criticizes other assignments of data from neighboring countries (e.g. in Asia)
- ▶ AJR cite sources which support assignment
- ▶ Albouy's results also depend on his "preferred" sample

Albouy:

Control variables	No controls (1)	Latitude control (2)	Without Neo- Europes (3)	Continent indicators (4)	Continent indicators and latitude (5)	Percent European in 1975 (6)	Malaria in 1994 (7)
<i>Panel B. Removing conjectured mortality rates (28 countries and mortality rates)</i>							
Log mortality (β)	-0.59	-0.42	-0.32	-0.31	-0.22	-0.29	-0.38
(heteroscedastic standard error)	(0.19)	(0.22)	(0.19)	(0.20)	(0.23)	(0.21)	(0.24)
<i>p</i> -value of log mortality	0.01	0.07	0.10	0.13	0.35	0.19	0.12
<i>p</i> -value of controls	—	0.05	—	0.01	0.002	0.015	0.10

Figure 6: First Stage, Dropping 36 Observations (Albouy 2012)

	Alternative series for settler mortality							
	Original AJR series	Original AJR series, capped at 250	Benchmarking to Caribbean	Benchmarking to Caribbean, capped at 250	Using naval stations, method 1	Using naval stations, method 1, capped at 250	Using naval stations, method 2	Using naval stations, method 2, capped at 250
	Dependent variable is protection against risk of expropriation							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
No covariates	-0.61	-0.94	-0.59	-0.91	-0.54	-0.77	-0.58	-0.88
(standard error)	(0.15)	(0.16)	(0.15)	(0.17)	(0.14)	(0.16)	(0.15)	(0.16)
(clustered standard error)	(0.17)	(0.18)	(0.17)	(0.18)	(0.16)	(0.18)	(0.17)	(0.17)
Number of clusters	36	36	36	36	35	35	36	36
Observations	64	64	64	64	53	53	64	64
With latitude	-0.52	-0.86	-0.5	-0.83	-0.43	-0.66	-0.49	-0.79
(standard error)	(0.17)	(0.18)	(0.16)	(0.19)	(0.15)	(0.19)	(0.16)	(0.18)
(clustered standard error)	(0.19)	(0.20)	(0.19)	(0.20)	(0.17)	(0.20)	(0.18)	(0.20)
Number of clusters	36	36	36	36	35	35	36	36
Observations	64	64	64	64	53	53	64	64
Without neo-Europes	-0.4	-0.66	-0.38	-0.64	-0.35	-0.52	-0.38	-0.61
(standard error)	(0.15)	(0.18)	(0.15)	(0.18)	(0.13)	(0.16)	(0.14)	(0.18)
(clustered standard error)	(0.17)	(0.20)	(0.16)	(0.19)	(0.14)	(0.16)	(0.16)	(0.18)
Number of clusters	33	33	33	33	32	32	33	33
Observations	60	60	60	60	49	49	60	60

Figure 7: First Stage, Alternative Mortality Series (AJR 2012)

Soldiers in Barracks Versus Soldiers on Campaign

Albouy:

- ▶ Living conditions of soldiers in barracks and those on campaign differ substantially
- ▶ Clean water and adequate sewage disposal can lower mortality rates from waterborne diseases
- ▶ Adequate shelter and nutritious food protect against malaria and lower its mortality
- ▶ AJR don't differentiate between barracks and campaigning soldiers, Albouy constructs a dummy
- ▶ Construct dummy for when data come from (harshly treated) African laborers

Albouy:

Control variables	No controls (1)	Latitude control (2)	Without Neo- Europes (3)	Continent indicators (4)	Continent indicators and latitude (5)	Percent European in 1975 (6)	Malaria in 1994 (7)
<i>Panel C. Original data, adding campaign and laborer indicators (64 countries, 36 mortality rates)</i>							
Log mortality (β)	-0.45	-0.39	-0.31	-0.37	-0.30	-0.27	-0.36
(heteroscedastic-clustered SE)	(0.18)	(0.20)	(0.17)	(0.22)	(0.23)	(0.19)	(0.21)
<i>p</i> -value of log mortality	0.020	0.06	0.09	0.09	0.20	0.17	0.10
<i>p</i> -value of indicators	0.16	0.22	0.31	0.26	0.35	0.19	0.21
<i>p</i> -value of controls	—	0.27	—	0.75	0.66	0.02	0.41
<i>Panel D. Removing conjectured mortality and adding campaign and laborer indicators (28 countries and mortality rates)</i>							
Log mortality (β)	-0.35	-0.21	-0.18	-0.25	-0.14	-0.20	-0.22
(heteroscedastic standard error)	(0.22)	(0.25)	(0.22)	(0.23)	(0.26)	(0.23)	(0.26)
<i>p</i> -value of log mortality	0.12	0.42	0.42	0.28	0.60	0.39	0.40
<i>p</i> -value of indicators	0.03	0.06	0.08	0.34	0.44	0.14	0.07
<i>p</i> -value of controls	—	0.07	—	0.03	0.01	0.05	0.28

Figure 8: First Stage, Including Campaign and Laborer Dummies (Albouy 2012)

Soldiers in Barracks Versus Soldiers on Campaign

AJR:

- ▶ Impossible from data to distinguish peace and campaign times
- ▶ Discard data from major wars
- ▶ Sometimes soldiers on campaign were healthier than in barracks
 - ▶ Don't stay long in one place, so water doesn't get contaminated
 - ▶ Changing location can end yellow fever epidemic since the range of it is quite small (some hundred yards)
- ▶ Albouy's coding is inconsistent and selective
 - ▶ Adjust coding for New Zealand and Hong Kong from noncampaign to campaign (and more)
- ▶ Neither of the dummies are significant in first or second stage

	AJR mortality series, capped at 250; Albouy campaign dummy	AJR mortality series, capped at 250; Albouy campaign dummy	AJR mortality series, minimal correction to Albouy campaign dummy	AJR mortality series, capped at 250; minimal correction to Albouy campaign dummy	AJR mortality series, extended correction to Albouy campaign dummy	AJR mortality series, capped at 250; extended correction to Albouy campaign dummy	AJR mortality series, Albouy sample of 28; campaign dummy	
Dependent variable is protection against expropriation								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
No covariates	-0.45	-0.77	-0.52	-0.84	-0.6	-0.91	-0.35	-0.96
(standard error)	(0.17)	(0.21)	(0.17)	(0.19)	(0.15)	(0.17)	(0.22)	(0.26)
(clustered standard error)	(0.18)	(0.21)	(0.18)	(0.19)	(0.18)	(0.18)	(0.22)	(0.26)
Number of clusters	36	36	36	36	36	36	28	27
Observations	64	64	64	64	64	64	28	27
With latitude	-0.39	-0.72	-0.45	-0.79	-0.53	-0.86	-0.21	-0.86
(standard error)	(0.19)	(0.23)	(0.18)	(0.21)	(0.17)	(0.19)	(0.25)	(0.27)
(clustered standard error)	(0.20)	(0.23)	(0.20)	(0.21)	(0.19)	(0.19)	(0.25)	(0.27)
Number of clusters	36	36	36	36	36	36	28	27
Observations	64	64	64	64	64	64	28	27
Without neo-Europes	-0.31	-0.54	-0.33	-0.57	-0.39	-0.63	-0.18	-0.5
(standard error)	(0.16)	(0.22)	(0.16)	(0.20)	(0.16)	(0.20)	(0.22)	(0.25)
(clustered standard error)	(0.17)	(0.20)	(0.17)	(0.19)	(0.17)	(0.20)	(0.22)	(0.25)
Number of clusters	33	33	33	33	33	33	25	24
Observations	60	60	60	60	60	60	25	24

Figure 9: First Stage, Adjustments to Campaign Dummy (AJR 2012)

Clustered Standard Errors

Albouy:

- ▶ Since some countries share the same source of mortality rates, residuals are correlated and clustered standard errors should be used
- ▶ First stage is weak, so use inverted AR statistic which is robust to weak instruments in the second stage

Control variables	No controls (1)	Latitude control (2)	Without neo-Europes (3)	Continent indicators (4)	Continent indicators & latitude (5)	Percent European in 1975 (6)	Malaria in 1994 (7)
<i>Panel A. Original mortality (64 countries, 36 mortality rates)</i>							
Expropriation risk (α)	0.93	0.96	1.24	0.97	1.07	0.92	0.62
Wald 95% conf. region	[0.52, 1.34]	[0.42, 1.50]	[0.35, 2.14]	[0.25, 1.70]	[-0.01, 2.16]	[0.28, 1.56]	[0.23, 1.01]
AR "95%" conf. region	[0.66, 1.83]	[0.64, 2.39]	[0.73, 7.04]	[0.50, 9.02]	$(-\infty, -3.08]$ U [0.41, $+\infty$)	[0.51, 6.45]	[0.28, 1.88]
<i>Panel B. Removing conjectured mortality rates (28 countries and mortality rates)</i>							
Expropriation risk (α)	0.87	0.82	1.15	1.12	1.25	0.94	0.71
Wald 95% conf. region	[0.43, 1.31]	[0.13, 1.51]	[-0.10, 2.40]	[-0.17, 2.42]	[-1.18, 3.67]	[-0.33, 2.21]	[-0.53, 1.96]
AR "95%" conf. region	[0.58, 2.01]	$(-\infty, -7.92]$ U [0.38, $+\infty$)	$(-\infty, -5.14]$ U [0.49, $+\infty$)	$(-\infty, -2.25]$ U [0.37, $+\infty$)	$(-\infty, +\infty)$	$(-\infty, -0.94]$ U [0.27, $+\infty$)	$(-\infty, +\infty)$
<i>Panel C. Original data, adding campaign and laborer indicators (64 countries, 36 mortality rates)</i>							
Expropriation risk (α)	1.09	1.15	1.45	1.06	1.19	1.18	0.66
Wald 95% conf. region	[0.32, 1.87]	[0.12, 2.18]	[-0.01, 2.91]	[0.07, 2.05]	[-0.30, 2.67]	[-0.29, 2.66]	[-0.50, 1.81]
AR "95%" conf. region	[0.62, 5.07]	$(-\infty, -17.59]$ U U [0.60, $+\infty$)	$(-\infty, -8.05]$ U [0.69, $+\infty$)	$(-\infty, -3.28]$ U [0.45, $+\infty$)	$(-\infty, -0.67]$ U [0.29, $+\infty$)	$(-\infty, -1.67]$ U [0.44, $+\infty$)	$(-\infty, +\infty)$ $(-\infty, +\infty)$

Figure 10: Second Stage, Various Robustness Checks with AR Confidence Intervals (Albouy 2012)

Alternative series for settler mortality								
	Original AJR series	Original AJR series, capped at 250	Benchmarking to Caribbean	Benchmarking to Caribbean, capped at 250	Using naval stations, method 1	Using naval sta- tions, method 1, capped at 250	Using naval stations, method 2	Using naval stations, method 2, capped at 250
Dependent variable is log GDP per capita in 1995								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
No covariates	0.93	0.82	0.96	0.85	1.01	0.94	0.98	0.86
AR confidence interval	[0.68, 1.40]	[0.62, 1.14]	[0.71, 1.47]	[0.64, 1.20]	[0.74, 1.63]	[0.70, 1.40]	[0.72, 1.50]	[0.65, 1.23]
AR confidence interval, clustered	[0.66, 1.72]	[0.60, 1.19]	[0.69, 1.85]	[0.63, 1.29]	[0.72, 1.90]	[0.69, 1.50]	[0.70, 1.85]	[0.64, 1.31]
<i>F</i> -stat, first stage	23.34	35.55	22.06	33.53	18.26	24.52	21.95	32.36
<i>F</i> -stat, first stage, clustered	12.45	28.09	11.72	25.31	11.96	19.00	12.05	25.11
With latitude	0.96	0.79	1.01	0.85	1.07	0.96	1.03	0.87
AR confidence interval	[0.65, 1.78]	[0.55, 1.24]	[0.68, 1.94]	[0.59, 1.36]	[0.70, 2.44]	[0.65, 1.79]	[0.70, 1.99]	[0.61, 1.41]
AR confidence interval, clustered	[0.64, 2.49]	[0.55, 1.20]	[0.68, 2.86]	[0.59, 1.35]	[0.71, 3.44]	[0.67, 1.82]	[0.69, 2.90]	[0.62, 1.40]
<i>F</i> -stat, first stage	13.48	21.82	12.67	20.37	9.66	13.44	12.52	19.46
<i>F</i> -stat, first stage, clustered	7.30	19.26	6.89	17.14	6.10	10.37	6.93	16.32

Figure 11: Second Stage, Alternative Mortality Series (AJR 2012)

	Albouy sample of 28	Albouy sample of 28, without Gambia	Albouy sample of 28, mortality capped at 250	Albouy sample of 28, without Gambia, mortality capped at 250	Original AJR series, without contested observations in West and Central Africa	Original AJR series, without contested observations in West and Central Africa, mortality capped at 250
Dependent variable is protection against risk of expropriation						
	(1)	(2)	(3)	(4)	(5)	(6)
No covariates	0.87	0.74	0.83	0.75	0.87	0.77
AR confidence set	[0.57, 1.64]	[0.50, 1.12]	[0.56, 1.40]	[0.52, 1.13]	[0.63, 1.32]	[0.56, 1.08]
AR confidence set, clustered	[0.60, 1.82]	[0.55, 1.02]	[0.59, 1.35]	[0.56, 1.07]	[0.62, 1.62]	[0.57, 1.12]
<i>F</i> -stat, first stage	12.47	22.31	17.13	24.38	22.55	33.34
<i>F</i> -stat, first stage, clustered	9.24	24.16	15.63	24.28	12.30	29.22
With latitude	0.82	0.63	0.73	0.62	0.89	0.75
AR confidence set	[0.40, 5.79]	[0.32, 1.31]	[0.35, 2.54]	[0.30, 1.31]	[0.59, 1.56]	[0.51, 1.17]
AR confidence set, clustered	[0.42, 19.00]	[0.35, 0.97]	[0.30, 1.63]	[0.24, 0.98]	[0.60, 2.09]	[0.51, 1.14]
<i>F</i> -stat, first stage	4.93	10.85	6.44	10.26	14.96	22.56
<i>F</i> -stat, first stage, clustered	3.62	14.42	6.50	11.84	8.02	19.49

Figure 12: Second Stage, Reduced Sample (AJR 2012)

AJR mortality series, capped at 250; Albouy preferred sample; extended correction to Albouy campaign dummy; dropping Gambia

	AJR mortality series, minimal correction to Albouy campaign dummy	AJR mortality series, capped at 250; Albouy campaign dummy	AJR mortality series, minimal correction to Albouy campaign dummy	AJR mortality series, capped at 250; minimal correction to Albouy campaign dummy	AJR mortality series, extended correction to Albouy campaign dummy	AJR mortality series, capped at 250; extended correction to Albouy campaign dummy	AJR mortality series, Albouy preferred sample; campaign dummy	
Dependent variable is log GDP per capita in 1995								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
No covariates	1.09	0.86	1.01	0.84	0.93	0.80	1.02	0.83
AR confidence set	[0.69, 2.61]	[0.56, 1.54]	[0.69, 1.85]	[0.58, 1.31]	[0.67, 1.45]	[0.59, 1.16]	$[-\infty, -4.58] \cup [0.42, \infty]$	[0.53, 1.44]
AR confidence set, clustered	[0.65, 3.96]	[0.51, 1.54]	[0.65, 2.52] $\cup$	[0.56, 1.35]	[0.65, 1.78]	[0.59, 1.18]	$[-\infty, -4.21] \cup [0.44, \infty]$	[0.58, 1.34]
F-stat, first stage	9.21	15.17	13.89	21.65	20.91	30.53	3.11	15.17
F-stat, first stage, clustered	5.9	13.78	8.02	19.2	11.49	26.27	2.57	13.17
With latitude	1.15	0.85	1.06	0.82	0.96	0.79	0.9	0.66
AR confidence set	[0.66, 4.87]	[0.50, 1.81]	[0.66, 2.64]	[0.52, 1.48]	[0.64, 1.76]	[0.54, 1.25]	$[-\infty, \infty]$	[0.33, 1.49]
AR confidence set, clustered	[0.60, 34.78]	[0.41, 1.69]	[0.62, 4.35]	[0.48, 1.41]	[0.64, 2.30]	[0.55, 1.18]	$[-\infty, \infty]$	[0.28, 1.08]
F-stat, first stage	5.91	10.52	8.69	14.61	13.81	21.09	0.89	8.94
F-stat, first stage, clustered	3.67	9.43	5.31	13.71	7.82	19.7	0.67	9.83

Figure 13: Second Stage, Modified Campaign Dummy (AJR 2012)

My Personal and Very Subjective Takeaways...

- ▶ Historical data is often low-quality, you have to make judgement calls
 - ▶ And there will be always someone who disagrees
- ▶ AJR's data is probably too low quality, be aware if you want to use it
- ▶ Comments that are only about the data are a bit annoying, if you want to criticize a paper make it about the methodology so that others can benefit from it
- ▶ Clustered standard errors are a good point
- ▶ Inverted AR statistic theoretically useful, but in practice difficult to interpret when confidence intervals are disconnected

References

Acemoglu, D., Johnson, S., & Robinson, J. A. (2001). The colonial origins of comparative development: An empirical investigation. *American economic review*, 91(5), 1369-1401.

Robinson, J. A., Johnson, S., & Acemoglu, D. (2005). A Response to Albouy's "A Reexamination Based on Improved Settler Mortality Data" *Unpublished*.

Robinson, J. A., Johnson, S., & Acemoglu, D. (2006). Reply to the Revised (May 2006) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.' *Unpublished*.

Robinson, J. A., Johnson, S., & Acemoglu, D. (2008). Reply to the Revised (2008) Version of David Albouy's 'The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data.' *Unpublished*.

Acemoglu, D., Johnson, S., & Robinson, J. A. (2012). The colonial origins of comparative development: An empirical investigation: Reply. *American Economic Review*, 102(6), 3077-3110.

Albouy, D. Y. (2004a). The Colonial Origins of Comparative Development: A Reinvestigation of the Data. *Unpublished*.

Albouy, D. Y. (2004b). The Colonial Origins of Comparative Development: A Reexamination Based on Improved Settler Mortality Data. *Unpublished*.

Albouy, D. Y. (2006). The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data. *Unpublished*.

Albouy, D. Y. (2008). The Colonial Origins of Comparative Development: An Investigation of the Settler Mortality Data. *Unpublished*.

Albouy, D. Y. (2012). The colonial origins of comparative development: an empirical investigation: comment. *American Economic Review*, 102(6), 3059-76.

Young, C. (1994). The African colonial state in comparative perspective. *Yale University Press*.